

Question 1

Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some of or all of the water in a jug into another jug.

The current state of the jugs will be described using notation $\text{jugwater} = (8, 0, 0)$, where jugwater_1 is the number of gallons of water in the 8-gallon jug, jugwater_2 is the number of gallons of water in the 5-gallon jug, and jugwater_3 is the number of gallons of water in the 3-gallon jug.

1. Pour the 8-gallon jug into the 3-gallon jug until it is full → $\text{jugwater} = (5, 0, 3)$
2. Empty the 3-gallon jug into the 5-gallon jug → $\text{jugwater} = (5, 3, 0)$
3. Pour the 8-gallon jug into the 3-gallon jug until it is full → $\text{jugwater} = (2, 3, 3)$
4. Pour the 3-gallon jug into the 5-gallon jug until it is full → $\text{jugwater} = (2, 5, 1)$
5. Empty the 5-gallon jug into the 8-gallon jug → $\text{jugwater} = (7, 0, 1)$
6. Empty the 3-gallon jug into the 5-gallon jug → $\text{jugwater} = (7, 1, 0)$
7. Pour the 8-gallon jug into the 3-gallon jug until it is full → $\text{jugwater} = (4, 1, 3)$
8. Now there is 4 gallons in the 8-gallon jug → it is possible to measure 4 gallons if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons only by successively pouring some of or all of the water in a jug into another jug.

Alternatively, you can place the 3-gallon jug inside the 5-gallon jug, such that the remaining volume is 2-gallons. Then fill that remaining space (in the 5-gallon jug) once, pour that inside the 3-gallon jug, place the 3-gallon jug back into the 5-gallon jug and fill the remaining space again, then pour the 3-gallon jug into the 5-gallon jug, giving 4 gallons in both the 5-gallon and 8-gallon jugs.

Question 2

Verify the $3x + 1$ conjecture for these integers.

The $3x + 1$ Conjecture: Let T be the transformation defined by $T(n) = 3n + 1$ if n is odd and $T(n) = n/2$ if n is even. Then for any positive integer n , the sequence $n, T(n), T(T(n)), T(T(T(n))), \dots$ eventually reaches 1.

a) 6

1. $T(6) = \frac{6}{2} = 3$
2. $T(3) = 3 \times 3 + 1 = 10$
3. $T(10) = \frac{10}{2} = 5$
4. $T(5) = 3 \times 5 + 1 = 16$
5. $T(16) = \frac{16}{2} = 8$
6. $T(8) = \frac{8}{2} = 4$
7. $T(4) = \frac{4}{2} = 2$
8. $T(2) = \frac{2}{2} = 1$
9. Thus, the sequence reaches 1.

b) 7

1. $T(7) = 3 \times 7 + 1 = 22$
2. $T(22) = \frac{22}{2} = 11$
3. $T(11) = 3 \times 11 + 1 = 34$
4. $T(34) = \frac{34}{2} = 17$
5. $T(17) = 3 \times 17 + 1 = 52$
6. $T(52) = \frac{52}{2} = 26$
7. $T(26) = \frac{26}{2} = 13$
8. $T(13) = 3 \times 13 + 1 = 40$
9. $T(40) = \frac{40}{2} = 20$
10. $T(20) = \frac{20}{2} = 10$
11. $T(10) = \frac{10}{2} = 5$
12. $T(5) = 3 \times 5 + 1 = 16$
13. $T(16) = \frac{16}{2} = 8$
14. $T(8) = \frac{8}{2} = 4$
15. $T(4) = \frac{4}{2} = 2$
16. $T(2) = \frac{2}{2} = 1$
17. Thus, the sequence reaches 1.

c) 17

1. $T(17) = 3 \times 17 + 1 = 52$
2. $T(52) = \frac{52}{2} = 26$
3. $T(26) = \frac{26}{2} = 13$
4. $T(13) = 3 \times 13 + 1 = 40$
5. $T(40) = \frac{40}{2} = 20$
6. $T(20) = \frac{20}{2} = 10$
7. $T(10) = \frac{10}{2} = 5$
8. $T(5) = 3 \times 5 + 1 = 16$
9. $T(16) = \frac{16}{2} = 8$
10. $T(8) = \frac{8}{2} = 4$
11. $T(4) = \frac{4}{2} = 2$
12. $T(2) = \frac{2}{2} = 1$
13. Thus, the sequence reaches 1.

d) 21

1. $T(21) = 3 \times 21 + 1 = 64$
2. $T(64) = \frac{64}{2} = 32$
3. $T(32) = \frac{32}{2} = 16$
4. $T(16) = \frac{16}{2} = 8$
5. $T(8) = \frac{8}{2} = 4$
6. $T(4) = \frac{4}{2} = 2$
7. $T(2) = \frac{2}{2} = 1$
8. Thus, the sequence reaches 1.

Question 3

Prove that you can use dominoes to tile a rectangular checkerboard with an even number of squares.

First, we show that if a checkerboard is rectangular with an even number of squares, then at least one side length is even.

The area of a rectangle is given by the formula $area = length \times width$.

For the sake of contrapositive, assume that both the length and width are odd.

Then there exists integers m and n such that $length = 2m + 1$ and $width = 2n + 1$.

Then the area of the rectangle is given by $area = (2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$.

Since integers are closed under addition and multiplication, $2mn + m + n$ is an integer. Therefore $2(2mn + m + n)$ is even and thus $2(2mn + m + n) + 1$ is odd.

Therefore, if both the length and width are odd, the area of the rectangle is odd. By contrapositive, if the area of the rectangle is even, then at least one side length is even.

Since at least one side is even, at least one side is divisible by 2.

Thus, we can take the even side and divide it into rows of height 2. Since these rows are of height 2, we can place dominoes vertically in each column of each row.

Therefore, if the board is rectangular with an even number of squares, we can tile the board with dominoes. ■

Question 4

Show that by removing two white squares and two black squares from an 8×8 checkerboard (assume only black and white colors are used) you can make it impossible to tile the remaining squares using dominoes.

If we remove two white squares and two black squares, we can isolate a single square such that it is effectively removed from the board. Since the rules state that dominoes must cover two squares to be placed, if any given tile has no adjacent tiles, it cannot be covered by a domino.

Thus, if we remove two white squares in, for example, positions (2, 2) and (2, 4) and two black squares in positions (1, 2) and (2, 1), then the white tile in position (1, 1) will have no adjacent tile and thus cannot be covered by a domino. At that point the board will essentially have 59 squares (64 initially, 60 after removing 2 white and 2 black, and 59 considering the unreachable tile at position (1, 1)).

Since 59 is an odd number, it is impossible to tile the board with dominoes, given that they are 1×2 in size and therefore require an even number of squares to tile.

Therefore, by removing two white squares and two black squares from an 8×8 checkerboard, we can make it impossible to tile the remaining squares using dominoes. ■

Question 5

Find these terms of the sequence $\{a_n\}$, where $a_n = 2(-3)^n + 5^n$.

a) a_0 . $a_0 = 2(-3)^0 + 5^0 = 2(1) + 1 = 2 + 1 = 3$

b) a_1 . $a_1 = 2(-3)^1 + 5^1 = 2(-3) + 5 = -6 + 5 = -1$

c) a_4 . $a_4 = 2(-3)^4 + 5^4 = 2(81) + 625 = 162 + 625 = 787$

d) a_5 . $a_5 = 2(-3)^5 + 5^5 = 2(-243) + 3125 = -486 + 3125 = 2639$

Question 6

What is the term a_8 of the sequence $\{a_n\}$, if a_n equals

a) 2^{n-1} . $a_8 = 2^{8-1} = 2^7 = 128$

b) 7 . $a_8 = 7$

c) $1 + (-1)^n$. $a_8 = 1 + (-1)^8 = 1 + 1 = 2$

d) $-(-2)^n$. $a_8 = -(-2)^8 = -256$

Question 7

What are the terms a_0, a_1, a_2, a_3 of the sequence $\{a_n\}$, where a_n equals

a) $2^n + 1$

- $a_0 = 2^0 + 1 = 1 + 1 = 2$
- $a_1 = 2^1 + 1 = 2 + 1 = 3$
- $a_2 = 2^2 + 1 = 4 + 1 = 5$
- $a_3 = 2^3 + 1 = 8 + 1 = 9$

b) $(n + 1)^{n+1}$

- $a_0 = (0 + 1)^{0+1} = 1^1 = 1$
- $a_1 = (1 + 1)^{1+1} = 2^2 = 4$
- $a_2 = (2 + 1)^{2+1} = 3^3 = 27$
- $a_3 = (3 + 1)^{3+1} = 4^4 = 256$

c) $\lfloor \frac{n}{2} \rfloor$

- $a_0 = \lfloor \frac{0}{2} \rfloor = 0$
- $a_1 = \lfloor \frac{1}{2} \rfloor = 0$
- $a_2 = \lfloor \frac{2}{2} \rfloor = 1$
- $a_3 = \lfloor \frac{3}{2} \rfloor = 1$

d) $\lfloor \frac{n}{2} \rfloor + \lceil \frac{n}{2} \rceil$

- $a_0 = \lfloor \frac{0}{2} \rfloor + \lceil \frac{0}{2} \rceil = 0 + 0 = 0$
- $a_1 = \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil = 0 + 1 = 1$
- $a_2 = \lfloor \frac{2}{2} \rfloor + \lceil \frac{2}{2} \rceil = 1 + 1 = 2$
- $a_3 = \lfloor \frac{3}{2} \rfloor + \lceil \frac{3}{2} \rceil = 1 + 2 = 3$

Question 8

List the first 10 terms of each of these sequences.

a) the sequence that begins with 2 and in which each successive term is 3 more than the preceding term.

2, 5, 8, 11, 14, 17, 20, 23, 26, 29

b) the sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms.

2, 4, 6, 10, 16, 26, 42, 68, 110, 178

Question 9

Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.

a) $a_n = 6a_{n-1}, a_0 = 2.$ 2, 12, 72, 432, 2592

b) $a_n = a_{n-1}^2, a_1 = 2.$ 2, 4, 16, 256, 65536

c) $a_n = a_{n-1} + 3a_{n-2}, a_0 = 1, a_1 = 2.$ 1, 2, 5, 11, 26

d) $a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1.$ 1, 1, 6, 27, 204

e) $a_n = a_{n-1} + a_{n-3}, a_0 = 1, a_1 = 2, a_2 = 0.$ 1, 2, 0, 1, 3

Question 10

Let $a_n = 2^n + 5 \cdot 3^n$ for $n = 0, 1, 2, \dots$

a) Find a_0, a_1, a_2, a_3, a_4 .

- $a_0 = 2^0 + 5 \cdot 3^0 = 1 + 5 = 6$
- $a_1 = 2^1 + 5 \cdot 3^1 = 2 + 15 = 17$
- $a_2 = 2^2 + 5 \cdot 3^2 = 4 + 45 = 49$
- $a_3 = 2^3 + 5 \cdot 3^3 = 8 + 135 = 143$
- $a_4 = 2^4 + 5 \cdot 3^4 = 16 + 405 = 421$

b) Show that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$, **and** $a_4 = 5a_3 - 6a_2$.

- $a_2 = 5a_1 - 6a_0$
1. $a_2 = 2^2 + 5 \cdot 3^2 = 4 + 45 = 49$
 2. $5a_1 - 6a_0 = 5(2) - 6(1) = 10 - 6 = 4$
 3. $a_2 = 5a_1 - 6a_0$

$$a_3 = 5a_2 - 6a_1$$

1. $a_3 = 2^3 + 5 \cdot 3^3 = 8 + 135 = 143$
2. $5a_2 - 6a_1 = 5(49) - 6(17) = 245 - 102 = 143$
3. $a_3 = 5a_2 - 6a_1$

$$a_4 = 5a_3 - 6a_2$$

1. $a_4 = 2^4 + 5 \cdot 3^4 = 16 + 405 = 421$
2. $5a_3 - 6a_2 = 5(143) - 6(49) = 715 - 294 = 421$
3. $a_4 = 5a_3 - 6a_2$

c) Show that $a_n = 5a_{n-1} - 6a_{n-2}$ **for** $n = 2, 3, 4, \dots$. For any $n \geq 2$, we have $a_n = 5a_{n-1} - 6a_{n-2}$.

Since $a_n = 2^n + 5 \cdot 3^n$, we have $a_{n-1} = 2^{n-1} + 5 \cdot 3^{n-1}$ and $a_{n-2} = 2^{n-2} + 5 \cdot 3^{n-2}$.

$$a_{n-1} = 2^{n-1} + 5 \cdot 3^{n-1} \text{ can be rewritten as } a_{n-1} = 2^{-1} \cdot 2^n + 5 \cdot 3^{-1} \cdot 3^n = \frac{1}{2} \cdot 2^n + \frac{5}{3} \cdot 3^n.$$

$$a_{n-2} = 2^{n-2} + 5 \cdot 3^{n-2} \text{ can be rewritten as } a_{n-2} = 2^{-2} \cdot 2^n + 5 \cdot 3^{-2} \cdot 3^n = \frac{1}{4} \cdot 2^n + \frac{5}{9} \cdot 3^n.$$

$$\text{Then } 5a_{n-1} - 6a_{n-2} \text{ is } 5\left(\frac{1}{2} \cdot 2^n + \frac{5}{3} \cdot 3^n\right) - 6\left(\frac{1}{4} \cdot 2^n + \frac{5}{9} \cdot 3^n\right) = \frac{5}{2} \cdot 2^n + \frac{25}{3} \cdot 3^n - \frac{3}{2} \cdot 2^n - \frac{10}{3} \cdot 3^n = 2^n + 5 \cdot 3^n = a_n.$$

Therefore, $a_n = 5a_{n-1} - 6a_{n-2}$ for $n = 2, 3, 4, \dots$ ■

Question 11

Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

a) $a_n = 0$. If $a_n = 0$, then $-3a_{n-1} + 4a_{n-2} = -3(0) + 4(0) = 0 = a_n$.

Therefore, $a_n = 0$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$. ■

b) $a_n = 1$. If $a_n = 1$, then $-3a_{n-1} + 4a_{n-2} = -3(1) + 4(1) = 1 = a_n$.

Therefore, $a_n = 1$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$. ■

c) $a_n = (-4)^n$. If $a_n = (-4)^n$, then $-3a_{n-1} + 4a_{n-2} = -3(-4)^{n-1} + 4(-4)^{n-2} =$

$$= \frac{-3}{4} \cdot (-4)^n + \frac{4}{16} \cdot (-4)^n = \frac{3}{4} \cdot (-4)^n + \frac{1}{4} \cdot (-4)^n = (-4)^n = a_n.$$

Therefore, $a_n = (-4)^n$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$. ■

d) $a_n = 2(-4)^n + 3$. If $a_n = 2(-4)^n + 3$, then $-3a_{n-1} + 4a_{n-2} = -3(2(-4)^{n-1} + 3) + 4(2(-4)^{n-2} + 3) =$

$$= \left(\frac{-6}{4} \cdot (-4)^n - 9\right) + \left(\frac{8}{16} \cdot (-4)^n + 12\right) = \frac{6}{4} \cdot (-4)^n - 9 + \frac{1}{2} \cdot (-4)^n + 12 = 2(-4)^n + 3 = a_n.$$

Therefore, $a_n = 2(-4)^n + 3$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$. ■

Question 12

Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if

a) $a_n = 0$. If $a_n = 0$, then $8a_{n-1} - 16a_{n-2} = 8(0) - 16(0) = 0 = a_n$.

Therefore, $a_n = 0$ **is** a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$. ■

b) $a_n = 1$. If $a_n = 1$, then $8a_{n-1} - 16a_{n-2} = 8(1) - 16(1) = 8 - 16 = -8 \neq a_n$.

Therefore, $a_n = 1$ **is not** a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$. ■

c) $a_n = 2^n$. If $a_n = 2^n$, then $8a_{n-1} - 16a_{n-2} = 8(2^{n-1}) - 16(2^{n-2}) = 4 \cdot 2^n - 4 \cdot 2^n = 0 \neq a_n$.

Therefore, $a_n = 2^n$ **is not** a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$. ■

d) $a_n = 4^n$. If $a_n = 4^n$, then $8a_{n-1} - 16a_{n-2} = 8(4^{n-1}) - 16(4^{n-2}) = \frac{8}{4} \cdot 4^n - \frac{16}{16} \cdot 4^n = 2 \cdot 4^n - 4^n = 4^n = a_n$.

Therefore, $a_n = 4^n$ **is** a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$. ■

Question 13

For each of these sequences find a recurrence relation satisfied by this sequence. (The answers are not unique because there are infinitely many different recurrence relations satisfied by any sequence.)

a) $a_n = 3$. Since this is a constant sequence, we state it as the sequence $a_n = a_{n-1}$.

b) $a_n = 2n$. Since this is an arithmetic sequence, it can be stated as $a_n = a_{n-1} + 2$.

c) $a_n = 2n + 3$. Since this is an arithmetic sequence, it can be stated as $a_n = a_{n-1} + 2$.

d) $a_n = 5^n$. Since this is a geometric sequence, it can be stated as $a_n = 5a_{n-1}$.

Question 14

Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if

a) $a_n = -n + 2$. If $a_n = -n + 2$, then $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ becomes $-n + 2 = -(n-1) + 2 + 2(-(n-2) + 2) + 2(n) - 9 \Rightarrow -n + 2 = -n + 1 + 2 - 2n + 8 + 2n - 9 \Rightarrow -n + 2 = -n + 2$.

Therefore, $a_n = -n + 2$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$. ■

b) $a_n = 5(-1)^n - n + 2$. If $a_n = 5(-1)^n - n + 2$, then $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ becomes $5(-1)^n - n + 2 = 5(-1)^{n-1} - (n-1) + 2 + 2(5(-1)^{n-2} - (n-2) + 2) + 2n - 9 \Rightarrow 5(-1)^n - n + 2 = -5(-1)^{n-1} + (n-1) + 2 + 10(-1)^{n-2} - 2(n-2) + 4 + 2n - 9 \Rightarrow 5(-1)^n - n + 2 = 5(-1)^n - n + 2$.

Therefore, $a_n = 5(-1)^n - n + 2$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$. ■

c) $a_n = 3(-1)^n + 2^n - n + 2$. If $a_n = 3(-1)^n + 2^n - n + 2$, then $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ becomes $3(-1)^n + 2^n - n + 2 = 3(-1)^{n-1} + 2^{n-1} - (n-1) + 2 + 2(3(-1)^{n-2} + 2^{n-2} - (n-2) + 2) + 2n - 9 \Rightarrow 3(-1)^n + 2^n - n + 2 = -3(-1)^{n-1} - 2^{n-1} + (n-1) + 2 + 6(-1)^{n-2} + 4^{n-2} - 2(n-2) + 4 + 2n - 9 \Rightarrow 3(-1)^n + 2^n - n + 2 = 3(-1)^n + 2^n - n + 2$.

Therefore, $a_n = 3(-1)^n + 2^n - n + 2$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$. ■

d) $a_n = 7 \cdot 2^n - n + 2$. If $a_n = 7 \cdot 2^n - n + 2$, then $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ becomes $7 \cdot 2^n - n + 2 = 7 \cdot 2^{n-1} - (n-1) + 2 + 2(7 \cdot 2^{n-2} - (n-2) + 2) + 2n - 9 \Rightarrow 7 \cdot 2^n - n + 2 = 7 \cdot 2^{n-1} - n + 1 + 14 \cdot 2^{n-2} - 2(n-2) + 4 + 2n - 9 \Rightarrow 7 \cdot 2^n - n + 2 = 7 \cdot 2^n - n + 2$.

Therefore, $a_n = 7 \cdot 2^n - n + 2$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$. ■

Question 15

Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach such as that used in class.

a) $a_n = -a_{n-1}, a_0 = 5.$

1. $a_1 = -a_0 = -5$
2. $a_2 = -a_1 = 5$
3. $a_3 = -a_2 = -5$
4. $a_4 = -a_3 = 5$

The sequence is $5, -5, 5, -5, \dots$, which can be stated as $a_n = 5(-1)^n$.

b) $a_n = a_{n-1} + 3, a_0 = 1.$

1. $a_1 = a_0 + 3 = 1 + 3$
2. $a_2 = a_1 + 3 = 1 + 3 + 3$
3. $a_3 = a_2 + 3 = 1 + 3 + 3 + 3$
4. $a_4 = a_3 + 3 = 1 + 3 + 3 + 3 + 3$

Each term is a series with n 3's added to 1. Therefore, a solution is $a_n = 3n + 1$.

c) $a_n = a_{n-1} - n, a_0 = 4.$

1. $a_1 = a_0 - 1 = 4 - 1$
2. $a_2 = a_1 - 2 = 4 - 1 - 2$
3. $a_3 = a_2 - 3 = 4 - 1 - 2 - 3$
4. $a_4 = a_3 - 4 = 4 - 1 - 2 - 3 - 4$
5. $a_5 = a_4 - 5 = 4 - 1 - 2 - 3 - 4 - 5$
6. $a_6 = a_5 - 6 = 4 - 1 - 2 - 3 - 4 - 5 - 6$

Each term is a series with $\sum_{i=1}^n i$ subtracted from 4. Since $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ (by the formula for the sum of the first n natural numbers), a solution is $a_n = 4 - \frac{n(n+1)}{2}$.

d) $a_n = 2a_{n-1} - 3, a_0 = -1.$

1. $a_1 = 2a_0 - 3 = 2(-1) - 3$
2. $a_2 = 2a_1 - 3 = 2(2(-1) - 3) - 3$
3. $a_3 = 2a_2 - 3 = 2(2(2(-1) - 3) - 3) - 3$
4. $a_4 = 2a_3 - 3 = 2(2(2(2(-1) - 3) - 3) - 3) - 3$

Each term is a series with $2^{n+1}(-1) - 3 \sum_{i=0}^n 2^i$ subtracted from -1. Since $\sum_{i=0}^n 2^i = 2^{n+1} - 1$ (by the formula for the sum of a geometric series), a solution is $a_n = 2^{n+1}(-1) - 3(2^{n+1} - 1) - 1$.